

PRD: Course Retake & Failed Status

1. Overview

Admins on the platform repeatedly ask: *"Can we automatically reset progress for users who fail a course?"* Behind this request sit two real problems. First, there is no automatic recovery path — when a learner finishes a course below the pass score, an admin must manually intervene to give them another shot, which is untenable at scale for compliance training. Second, the progress display can feel contradictory — a learner can reach 100% content completion while still burning through per-question quiz retries, and admins need to understand that state without confusion.

This spec introduces three things: (1) an **info icon** that disambiguates the "100% complete but still retrying quizzes" state, (2) a **Reset progress** primitive (usable both automatically and manually) that gives learners a fresh course attempt *within the same enrolment* — preserving due dates, start dates, and compliance cycles, and (3) an **"Attempt no" column** and admin controls to surface repeat-strugglers. The design follows the dominant industry pattern: the enrolment is durable; progress data on it gets reset; past attempts are preserved in a history log.

2. Jobs To Be Done

Main Job

When a learner fails a course, give them another chance to pass without disrupting the compliance/training cycle or requiring manual admin intervention.

Job Map

Stage	What happens today	Pain / Workaround
Define	Admin sets up a course with a pass threshold and per-question retry limits	No ability to define what happens <i>after</i> failure — the system has no concept of "course attempt"
Locate	Admin identifies a failed learner on the enrolments table	Status model has no "Failed" state; a learner stuck below threshold looks identical to one still working — both show as "In Progress"
Prepare	Admin decides the learner needs another shot	No automated option; admin must mentally track who needs resetting

Confirm	Admin initiates a re-enrol (only existing action)	Re-enrol creates a <i>new</i> enrolment record, shifting due dates out of sync with the compliance calendar — wrong tool for the job
Execute	Re-enrol runs; learner starts on a new cycle	Compliance cycle is broken; learner is now on a personal failure-recovery schedule instead of the company's annual training calendar
Monitor	Admin checks if the learner passed on the second try	No "Attempt no" concept — admin cannot distinguish first-time learners from repeat-strugglers without memory or spreadsheets
Resolve	Compliance audit requires proof of attempts	No CourseAttemptHistory — previous attempt data is either lost (re-enrol) or invisible

Related Jobs

- **Compliance officer** needs an audit trail proving that a learner attempted a course, failed, and was given a structured opportunity to retry — all within the same assignment period.
- **Learner** needs to understand what happened (they didn't pass) and what to do next (try again) without feeling punished or confused.
- **Admin managing hundreds of enrolments** needs to quickly filter and find repeat-strugglers who may need coaching or alternative interventions, without reviewing each record individually.

Emotional & Social Dimensions

- **Admin** wants to feel *in control* and *efficient*. They don't want to be a bottleneck in the learner's recovery path. They want to trust the system to handle the common case automatically while surfacing the edge cases that need human judgment.
- **Learner** wants to feel that failure is *recoverable*, not *terminal*. The system should frame a new attempt as a fresh opportunity, not a mark of shame. Seeing "Failed" as a permanent end-state (when auto-reset is off) should be rare and clearly explained.
- **Compliance officer** wants to feel *confident* that the audit trail is complete and immutable — that the system's records will hold up under regulatory scrutiny.

3. Goals

1. **Eliminate manual admin intervention for the common case.** When auto-reset is enabled, a learner who fails should automatically get a fresh attempt — no admin action required.
2. **Preserve compliance cycle integrity.** Reset progress must never shift a learner's start date, due date, or recurrence cycle. The enrolment record is durable.

3. **Disambiguate the "100% complete but still retrying" state.** Admins should never be confused by a learner showing 100% progress but "In Progress" status.
 4. **Surface repeat-strugglers without inventing new status labels.** The "Attempt no" column + filter lets admins find learners on attempt 3, 4, 5+ for proactive coaching — without overloading the status taxonomy.
 5. **Maintain a complete audit trail.** Every closed attempt is snapshotted into CourseAttemptHistory before progress resets, and old QuestionAnswer records remain queryable by attempt number.
 6. **Keep the status model simple.** Three statuses only: In Progress, Passed, Failed. The info icon handles the single ambiguous case.
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4. Job Stories

Course Recovery

- **When** a learner finishes a course below the pass score with all quiz retries exhausted and auto-reset is on, **I want** the system to automatically reset their progress and open a new attempt, **so I can** avoid manually resetting dozens of failed learners in compliance training cycles.
- **When** a learner finishes a course below the pass score with all quiz retries exhausted and auto-reset is off, **I want** to see a clear "Failed" status on the enrolments table with a "Reset progress" action available, **so I can** decide whether to give them another shot or take a different intervention.
- **When** I click "Reset progress" on a learner (in any status), **I want** to see a confirmation modal that clearly states what will happen — progress archived, fresh attempt opened, dates unchanged — **so I can** be confident I'm not disrupting the enrolment cycle.

Status Clarity

- **When** I see a learner at 100% completion but still "In Progress," **I want** an info icon that explains they're working through remaining quiz retries, **so I can** understand the state without contacting support or inspecting raw data.
- **When** I'm reviewing the enrolments table for a compliance audit, **I want** to see an "Attempt no" column showing how many times each learner has attempted the course, **so I can** identify repeat-strugglers and decide if they need coaching or alternative training.

Learner Experience

- **When** my progress is auto-reset after a failed attempt, **I want** to receive a notification ("Your fresh attempt at [Course] is open") without first seeing a jarring "Failed" state, **so I can** feel that

the system is helping me recover rather than punishing me.

- **When** my progress is reset (auto or manual), **I want** to only need to retake the quizzes — not re-consume all the content (videos, readings) — **so I can** focus my time on what I actually need to improve.

Admin Configuration

- **When** I'm setting up a compliance course, **I want** a per-course "Auto-reset on failure" toggle (off by default), **so I can** decide on a course-by-course basis whether failed learners are automatically given another shot.
 - **When** I configure auto-reset, **I want** to also set a maximum number of course attempts, **so I can** prevent infinite retry loops and escalate persistent failures to human review.
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5. Requirements

Functional Requirements

FR-1: Status Model (Three statuses + info icon)

- **FR-1.1:** The system SHALL display exactly three admin-facing statuses: **In Progress**, **Passed**, **Failed**.
- **FR-1.2:** Status derivation after every quiz answer submission:
 - `quizResult === true` → **Passed**
 - `quizResult === false` → check quiz attempt exhaustion (FR-1.3)
 - `quizResult === undefined` → **In Progress**
- **FR-1.3:** When `quizResult === false` :
 - If any failing question still has remaining quiz attempts → **In Progress** (show info icon if at 100% content completion)
 - If all failing questions are at `maxAttempts` :
 - If `autoResetEnabled === true` → execute Reset progress; learner becomes **In Progress** (new attempt)
 - If `autoResetEnabled === false` → **Failed**
- **FR-1.4:** When a learner is at 100% content completion but status is In Progress (quiz retries remaining), a small info icon SHALL appear next to the status badge. Hovering (mouse or

keyboard) shows tooltip: "Content completed. Learner is working through remaining quiz retries to reach the pass score."

- **FR-1.5:** The info icon and tooltip are admin-facing only (shown on the enrolments table). The learner does not see an equivalent message.

FR-2: Course Attempt Concept

- **FR-2.1:** A new field `courseAttemptNumber` (integer, starting at 1) SHALL be added to the CourseEnrolment record.
- **FR-2.2:** `courseAttemptNumber` increments only when Reset progress runs (auto or manual). Per-question retries do NOT increment it.
- **FR-2.3:** A `courseAttemptNumber` field SHALL be added to QuestionAnswer records to scope answers to a specific course attempt, enabling reporting queries by attempt.

FR-3: Reset Progress Primitive

- **FR-3.1:** Reset progress is a single operation used by both auto-reset and the admin manual action. It SHALL:
 1. Snapshot the current enrolment state into a `CourseAttemptHistory` row (attempt number, score, completion date, close reason — `auto-reset` or `admin-reset` — and admin user ID if applicable). Per-question answers are referenced by `courseAttemptNumber` on the existing `courseEnrolmentId`, not duplicated.
 2. Reset progress fields on the existing enrolment: `progress: 0`, `quizScore: 0`, `quizResult: undefined`, `completionDate: null`, `passed: false`, `taken: false`.
 3. Increment `courseAttemptNumber` by 1.
 4. Leave `startDate`, `endDate`, `dueDate`, and recurrence untouched.
 5. Old QuestionAnswer records remain in the database, scoped by `courseAttemptNumber`; they are inactive for the new attempt.
 6. Emit an event hook for downstream listeners (notifications).
- **FR-3.2:** On reset, only quiz progress is reset. Learners do NOT need to re-consume content (videos, readings, etc.) — they retake quizzes only.

FR-4: Auto-Reset Trigger

- **FR-4.1:** Auto-reset fires when ALL four conditions are true:
 1. Course is at 100% content completion
 2. `quizResult === false`
 3. Every failing question has used all its per-question attempts (`attemptNumber === maxAttempts` for every question where `passed: false`)

4. Course has `autoResetEnabled: true`

- **FR-4.2:** If conditions 1–3 hold but `autoResetEnabled === false`, the learner moves to **Failed**.
- **FR-4.3:** No math-elimination check. The system does NOT compute whether remaining retries can mathematically reach the threshold. Only attempt counts are checked.
- **FR-4.4:** Auto-reset is immediate — no cooldown window.
- **FR-4.5:** When auto-reset fires, the transient "Failed" state SHALL be suppressed in the learner-facing UI. The learner's status transitions directly from In Progress → In Progress (new attempt) without ever displaying "Failed." The notification ("Your fresh attempt at [Course] is open") provides sufficient context.

FR-5: Per-Course Settings

- **FR-5.1:** A new per-course setting "**Auto-reset on failure**" (toggle, default: Off). When enabled, auto-reset fires per FR-4.
- **FR-5.2:** A new per-course setting "**Maximum course attempts**" (integer, admin-defined). When the learner reaches this limit, auto-reset does not fire regardless of `autoResetEnabled`; the learner moves to **Failed** and admin must manually intervene. No system-generated warning about attempt count is needed.

FR-6: Admin Enrolments Table

- **FR-6.1:** A new column "**Attempt no**" SHALL be added to the enrolments table, showing `courseAttemptNumber` for each enrolment.
- **FR-6.2:** The column is sortable.
- **FR-6.3:** The column is included in the learner CSV export.
- **FR-6.4:** (Phase 3) A filter on `Attempt no ≥ N`, combinable with the Status filter, helps admins find repeat-strugglers.

FR-7: Admin Row-Level "Reset Progress" Action

- **FR-7.1:** A "Reset progress" action is available on each learner's row in the enrolments table, for any status (In Progress, Passed, Failed).
- **FR-7.2:** It is distinct from the existing "Re-enrol" action.
- **FR-7.3:** Clicking it shows a confirmation modal: *"Reset this learner's progress on [Course]? Their current attempt will be archived and they'll start a fresh course attempt within the same enrolment. Their start date, due date, and recurrence cycle are unchanged."*
- **FR-7.4:** The confirmation modal uses descriptive button labels (e.g., "Reset Progress" / "Cancel"), not generic "Yes" / "No."
- **FR-7.5:** The audit log records `closeReason: 'admin-reset'` and the admin user ID, distinguishing it from `closeReason: 'auto-reset'`.

FR-8: Notifications (Phase 3)

- **FR-8.1:** When Reset progress fires (auto or manual), the learner receives an in-app + email notification: *"Your fresh attempt at [Course] is open."*

FR-9: Explicit Non-Requirements

- No new status labels (e.g., "Retaking," "Pass Pending").
 - No math-elimination detection mid-course.
 - No cooldown window before auto-reset.
 - No new enrolment record per retake.
 - No changes to the existing Re-enrol action.
 - No partial question resets (only resetting failed questions).
 - No learner-facing "Restart course" button.
 - No attempt history visibility for learners (deferred).
 - No detailed attempt history view for admins (Phase 2 shows current attempt number only; detailed history deferred).
 - No downstream/webhook consumers to conform to for the event hook.
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6. Research & Best Practices

Industry Pattern Validation

The design decisions in this spec are strongly validated by how mature LMS platforms handle the same problem. The core pattern — durable enrolment record with in-place progress reset and a separate attempt history log — is used by Cornerstone, Docebo, SAP Litmos, Absorb, TalentLMS, 360Learning, and Moodle.

Absorb LMS resets all lesson progress while archiving previous enrollments as "Historic Enrollments" with full reporting data preserved. Admin can configure auto-re-enrollment based on completion date or certificate expiry ([Re-Enrollment & Re-Certification — Absorb LMS Help Center](#)). Their automatic re-enrollment uses a two-step toggle pattern (enable + configure) that is a useful reference for the "Auto-reset on failure" course setting ([How to Automate Course Re-Enrollment — Absorb LMS](#)). Absorb also consolidates training data into centralized, audit-ready reports with cascading notification patterns (learner → manager → compliance team) that could inform future phases ([Automated Compliance Reporting — Absorb LMS](#)).

Docebo provides admin-configurable reset behavior; the default for compliance is in-place progress reset on the existing enrolment. Their community reveals a critical UX pitfall: the "Renew" button archives partial progress if clicked mid-attempt, leading to data loss. Community members advocate for a confirmation message like *"You have already re-enrolled during your enrollment period. Are you sure you want to re-enroll and reset your progress?"* — directly validating our confirmation modal approach ([Resetting Learner Progress — Docebo Community](#)). Docebo's audit trail is an immutable log managed by Superadmins, tracking all administrative actions including enrollment and completion changes ([Docebo Audit Trail Events](#)

[Details](#)), and their compliance solution replaces manual tracking with automated notifications, renewal cycles, and dashboards ([How to Automate Compliance Training — Docebo](#)).

Moodle's "Interactive with multiple tries" quiz mode is the closest analog to the per-question retry mechanism already in place. Moodle resets completion state in-place with quiz attempts logged separately ([Quiz Settings — MoodleDocs](#)). Notably, Moodle has a known bug where quiz activities with "complete on passing grade" incorrectly unlock content even when a student fails — underscoring why our status derivation logic (checking `quizResult` rather than completion percentage) is the correct approach ([Fixing Moodle's Quiz Completion Bug — InfraNext](#)). Moodle's dev team also documents the compliance danger of retroactively changing completion states — a student might have filed a certificate with regulators, and changing pass thresholds later could invalidate it — reinforcing our approach of never modifying the closed-attempt record ([Policy: Retroactive Effects of Completion Settings — MoodleDocs](#)).

Canvas community discussions reveal that the "retake after failure" workflow is one of the most-requested and under-served features across LMS platforms ([How to Implement Course Retake — Canvas Community](#)). **LearnDash** requires a third-party add-on for progress resets, and without a separate completion-records plugin, historical completion data is destroyed on reset — validating our decision to snapshot into CourseAttemptHistory before resetting ([Automatically Reset LearnDash Courses — WooNinjas](#); [Complete Guide to Resetting Student Progress in LearnDash — WooNinjas](#)).

A cross-platform review confirms that automated re-enrollment/reset, audit trails, and admin notifications are table-stakes features for compliance LMS platforms in 2025–2026 ([10 Best Compliance Training Software Solutions — Docebo](#)).

Status Anti-Pattern: "Stuck In Progress"

A well-documented anti-pattern occurs when a learner fails a course but the status gets "stuck" as "In Progress" or "Resume" due to poor communication between the SCORM package and LMS on session state after failure ([Course Stuck in Resume/In-Progress — Adobe Community](#)). Our info-icon approach directly addresses this: when a learner *is* legitimately "In Progress" at 100% (still has quiz retries), the icon disambiguates the state; when they've exhausted retries, they transition cleanly to Failed or auto-reset.

Transient State Suppression (Answer to Q2)

Industry best practice and UX research converge on suppressing the transient "Failed" flash when auto-reset is enabled:

- Error state design patterns recommend that backend workflows should "tolerate minor hiccups, retry failed steps, and gracefully degrade without blocking the user journey" — then surface friendly messages at completion ([Error State Design Patterns — 2Point Agency](#)).
- Modal windows should be reserved for errors that are "super destructive or cause major blocks in the workflow." For transient states that self-resolve (like auto-reset), a non-blocking notification is more appropriate ([Error Handling UX Design Patterns — Medium/Bootcamp](#)).
- The retry pattern in microservices establishes the principle that retryable conditions should not be surfaced as terminal failures ([Retry Pattern in Microservices — GeeksforGeeks](#)).

The learner should never see "Failed" when auto-reset is on. The status transitions directly from In Progress → In Progress (new attempt), with a notification providing context.

Confirmation Modal Best Practices

NNg's 8 principles for destructive-action confirmations validate the spec's approach: (1) reserve for serious consequences, (2) avoid overuse, (3) provide specificity about what will happen, (4) use descriptive button labels (not "Yes/No"), (5) enable progressive disclosure, (6) avoid default answers. The spec's confirmation copy ("Reset this learner's progress on [Course]? Their current attempt will be archived and they'll start a fresh course attempt within the same enrolment. Their start date, due date, and recurrence cycle are unchanged.") follows principles 1, 3, and 4 well ([Confirmation Dialogs — NN/g](#)).

Tooltip and Indicator Pattern

NNg's tooltip guidelines specify that tooltips should never contain information vital to task completion; they're best for supplementary context, must trigger via both mouse and keyboard hover, and should be kept as microcontent ([Tooltip Guidelines — NN/g](#)). The info-icon tooltip ("Content completed. Learner is working through remaining quiz retries to reach the pass score.") aligns with this — it's supplementary disambiguation, not a required action prompt.

NNg's framework distinguishes three communication methods: indicators (passive, contextual icons), validations (error messages), and notifications (system alerts). The info icon is correctly classified as an indicator; the auto-reset notification to learners is correctly a notification ([Indicators, Validations, and Notifications — NN/g](#)).

Mobile & UX Context

LMS UI/UX design directly influences knowledge retention rates (up to 60% with custom corporate LMS), and retake flows must work seamlessly across devices given 13% CAGR mobile learning growth through 2026 ([LMS UI/UX Design — Riseapps](#)). The info icon and tooltip should be tested for touch devices (long-press trigger) and small screens.

UX References

App	Flow / Screen	Mobbin URL	Pattern Description	Relevance
Teachable	Admin enrollment/student list	View	Course enrollment table with status columns and student data	Direct competitor — reference for how the enrolments table structures status, student names, and progress columns
Teachable	Course enrollment detail	View	Enrollment data table with column headers and row actions	Reference for "Attempt no" column placement alongside existing columns

Teachable	Course settings panel	View	Admin course configuration with toggles	Direct reference for "Auto-reset on failure" toggle placement in course settings
Teachable	Course admin settings	View	Additional course configuration options	Shows Teachable's pattern for organizing course-level settings and completion rules
Teachable	Course enrollment listing	View	Student enrollment table with filters	Reference for filter UI — relevant to "Attempt no $\geq$ N" filter in Phase 3
Teachable	Course completion settings	View	Completion and certificate configuration	Shows where auto-reset toggle would live alongside completion/certification settings
Deel	Admin enrollment/user table	View	HR data table with status column and row actions	Shows how a SaaS admin table handles status badges with contextual actions per row
Deel	User management detail	View	Admin detail view with status indicators	Reference for status badge + info icon pattern on individual records
Gusto	Admin enrollment/status table	View	Admin table with enrollment status column	Shows how an HR/admin SaaS structures status columns in data tables
Fresha	Status badge with tooltip	View	Info tooltip on status indicator in table	Directly relevant — shows the info-icon-next-to-status-badge pattern for disambiguating states
Airtable	Status badge with contextual info	View	Status badge with supplementary info in data table	Reference for adding contextual information alongside status badges without cluttering the table
Remote	Status indicator with tooltip	View	Status column with info icon tooltip	HR SaaS showing status disambiguation via hover — directly relevant to the info icon at 100% In Progress
Remote	Admin table with actions	View	Row-level actions in admin data table	Reference for how "Reset progress" row action should be positioned alongside other actions

Neon	Destructive action confirmation modal	View	Confirmation dialog for irreversible action	Directly relevant to the "Reset progress" confirmation modal — shows copy and button patterns for destructive confirmations
GoDaddy	Destructive action confirmation	View	Confirmation dialog with specific consequence description	Reference for describing consequences clearly in the reset confirmation modal
Medium	Content action confirmation	View	Confirmation modal with descriptive buttons	Shows Medium's approach to descriptive button labels per NNg guidelines
Podia	Course settings with toggles	View	Course admin settings panel with toggle switches	Direct competitor — reference for toggle-based course settings layout
Squarespace	Settings configuration panel	View	Admin settings page with toggle configurations	Reference for layout of auto-reset setting alongside other configuration
Circle	Course/community settings	View	Admin configuration panel for content settings	Shows how a learning platform structures per-course admin settings

Built for Mars Lessons

Title	Type	URL	Lesson	Relevance
Benchmarking the UX of 6 Delivery Companies	Case study	View	Confusing or erroring status indicators are "damaging" to user trust. Status tracking must be unambiguous even for non-terminal states. When a temporary status is off-screen, users crave a place to store details and track progression.	Directly validates the info icon on "In Progress" at 100% — disambiguating a non-terminal but potentially confusing state prevents the trust erosion BFM identifies with unclear statuses.
The Art (and Science) of Creating Goals	Case study	View	The best retention strategy helps users feel like they're making progress from the moment they set a goal.	Relevant to the course attempt counter and progress reset — resetting progress should frame the new attempt as a

That Actually Stick (Strava)			Status management must handle users falling behind or recovering momentum through thoughtful interface design.	fresh opportunity, not a punishment, maintaining the learner's sense of momentum.
Cognitive Load	Glossary	View	Reducing cognitive load is essential in admin interfaces — using existing, understood labels plus small indicators rather than inventing new concepts directly reduces the interpretation burden.	Validates the decision against creating new status labels like "Retaking" or "Pass Pending" — each new concept increases cognitive load for admins scanning potentially hundreds of enrollments.
Decision Fatigue	Glossary	View	Every decision point in an admin workflow costs cognitive energy. Automating the most common response to a condition reduces decision fatigue and preserves attention for edge cases.	Supports the auto-reset feature — by automating the standard response to failure (reset and retry) for compliance courses, the spec reduces per-learner decisions, preserving admin attention for the learners who actually need human intervention.

7. Plan of Action

Phase 1: Info Icon at 100% In Progress ([DEV-4347](#))

Lowest-risk change. No data model changes.

- ☐ Update status derivation logic in `progress.ts` / `helpers.ts` to detect the "100% content complete + quiz retries remaining + `quizResult === false`" state
- ☐ Design and implement the info icon component next to the "In Progress" status badge (reference: Fresha, Remote, Airtable Mobbin patterns)
- ☐ Implement tooltip on hover (mouse + keyboard accessible): *"Content completed. Learner is working through remaining quiz retries to reach the pass score."*
- ☐ Ensure tooltip works on touch devices (long-press trigger)
- ☐ Add unit tests for the new status derivation branch
- ☐ QA: Verify info icon appears only in the correct state (100% completion, quiz retries remaining, `quizResult === false`)
- ☐ QA: Verify info icon does NOT appear when `quizResult === true` (Passed) or `quizResult === undefined` (still completing content)

Phase 2: Auto-Reset + Attempt Tracking + Admin Reset Action ([DEV-4348](#))

Core data model and feature changes.

- ☐ **Data model changes:**
 - ☐ Add `courseAttemptNumber` (integer, default 1) to `CourseEnrolment`
 - ☐ Add `courseAttemptNumber` field to `QuestionAnswer` records for attempt scoping
 - ☐ Create `CourseAttemptHistory` collection/table (fields: `courseEnrolmentId`, `attemptNumber`, `score`, `completionDate`, `closeReason`, `adminUserId`, `closedAt`)
 - ☐ Add `autoResetEnabled` (boolean, default false) to `CourseSettings`
 - ☐ Add `maxCourseAttempts` (integer, admin-defined) to `CourseSettings`
 - ☐ Write migration scripts; ensure backward compatibility (existing enrolments get `courseAttemptNumber: 1`)
- ☐ **Reset progress service:**
 - ☐ Implement the Reset progress primitive (snapshot → reset fields → increment attempt → preserve dates → emit event)
 - ☐ Ensure quiz-only reset: progress on content (videos, readings) is not reset; only quiz-related fields (`quizScore`, `quizResult`, `passed`, `taken`, per-question answers) are reset
 - ☐ Ensure old `QuestionAnswer` records remain queryable by `courseAttemptNumber`
 - ☐ Unit test: Reset preserves `startDate`, `endDate`, `dueDate`, recurrence
 - ☐ Unit test: Reset increments `courseAttemptNumber`
 - ☐ Unit test: `CourseAttemptHistory` row is created with correct fields
 - ☐ Unit test: `maxCourseAttempts` is respected — no reset when limit reached
- ☐ **Auto-reset trigger:**
 - ☐ Implement the four-condition check after every quiz answer submission
 - ☐ Suppress transient "Failed" state in learner-facing UI when auto-reset fires (status transitions directly to In Progress with new attempt number)
 - ☐ Unit test: Auto-reset fires when all four conditions met
 - ☐ Unit test: Auto-reset does NOT fire when `autoResetEnabled === false`

- ☐ Unit test: Auto-reset does NOT fire when `courseAttemptNumber >= maxCourseAttempts`
- ☐ Unit test: Learner moves to Failed when auto-reset is off and retries exhausted
- ☐ **Per-course settings UI:**
 - ☐ Add "Auto-reset on failure" toggle to course settings (reference: Teachable, Podia, Squarespace Mobbin patterns)
 - ☐ Add "Maximum course attempts" number input (visible when auto-reset is on, or always visible)
 - ☐ Default: Off / no limit
- ☐ **Admin enrolments table:**
 - ☐ Add "Attempt no" column displaying `courseAttemptNumber` (reference: Teachable enrollment table Mobbin patterns)
 - ☐ Make column sortable
 - ☐ Include in CSV export
 - ☐ Add "Reset progress" row-level action for all statuses (reference: Remote admin table Mobbin pattern)
 - ☐ Implement confirmation modal with descriptive copy and buttons (reference: Neon, GoDaddy, Medium Mobbin patterns; NNg confirmation dialog guidelines)
 - ☐ Record `closeReason` ('admin-reset') and admin user ID in `CourseAttemptHistory`
- ☐ **Integration tests:**
 - ☐ Happy path 1: Learner passes first try → no reset, attempt stays at 1
 - ☐ Happy path 2: Learner struggles with quiz retries, passes → no reset, info icon shown transiently at 100%
 - ☐ Non-happy path (auto-reset on): Failure → auto-reset → new attempt → learner passes
 - ☐ Non-happy path (auto-reset off): Failure → Failed status → admin manual reset → new attempt
 - ☐ Edge case: Admin resets a Passed learner → new attempt opens, quiz-only reset
 - ☐ Edge case: Learner reaches `maxCourseAttempts` → Failed, no auto-reset

Phase 3: Notifications & Admin Filter ([DEV-4354](#))

- ☐ Implement in-app notification for learners on Reset progress: *"Your fresh attempt at [Course] is open."*

- ☐ Implement email notification for learners on Reset progress (same copy)
- ☐ Notifications fire for both auto-reset and admin-triggered reset
- ☐ Add "Attempt no $\geq$ N" filter to the enrolments table, combinable with Status filter (reference: Teachable enrollment listing Mobbin pattern)
- ☐ QA: Verify filter + sort + CSV export all work together
- ☐ QA: Verify notifications are sent correctly for both trigger types

8. Risks & Mitigations

Risk	Impact	Mitigation
Infinite retry loops — a learner with auto-reset on and no max attempts could cycle indefinitely without learning	Wasted learner time; no compliance value; admin doesn't notice	Admin-defined <code>maxCourseAttempts</code> setting caps retries. "Attempt no" column + filter (Phase 3) surfaces repeat-strugglers for proactive coaching.
Data model migration breaks existing enrolments — adding <code>courseAttemptNumber</code> to existing records could cause null errors or reporting inconsistencies	Data integrity issues; broken reports	Migration sets <code>courseAttemptNumber: 1</code> for all existing enrolments. Backward-compatible: all existing code paths treat absence as 1.
QuestionAnswer scoping by attempt number introduces query complexity — reporting queries must now filter by <code>courseAttemptNumber</code> , and existing queries could return stale cross-attempt data	Incorrect reporting; compliance audit failures	Add <code>courseAttemptNumber</code> index; update all existing QuestionAnswer queries to scope to the current attempt; integration tests validate scoping.
Transient "Failed" state leaks to learner UI — race condition where the UI renders before auto-reset completes	Learner sees "Failed" briefly; anxiety and confusion	Status derivation should check auto-reset eligibility <i>before</i> writing "Failed" to the learner-facing state — if auto-reset conditions are met, skip the Failed state entirely and execute reset in the same transaction.

Admin resets a Passed learner by mistake — the action is available for all statuses, including Passed	Learner loses their passing record; compliance implications	Confirmation modal explicitly states consequences. CourseAttemptHistory preserves the Passed attempt record for audit. The modal copy could be enhanced for Passed learners: "This learner has already passed. Resetting will require them to retake quizzes."
Info icon not discoverable on mobile / touch devices — hover-based tooltips don't work natively on touch	Admins on tablets miss the disambiguation context	Implement long-press trigger for touch; consider making the tooltip text visible inline on small screens. Test across devices in Phase 1 QA.
Auto-reset fires on overdue enrolments — a learner past their due date fails and gets auto-reset, but the due date is already passed	Learner gets a new attempt they can't meaningfully complete before the deadline	Deferred per clarifying answers (edge case for now). Future consideration: suppress auto-reset when due date has passed, or warn admin.
Confirmation modal fatigue — admins batch-resetting many learners must click through a modal each time	Slowed admin workflow; frustration	Phase 1 scope is single-learner reset. Future: consider bulk-reset with a single confirmation ("Reset progress for N learners?") if admins request it.
Audit trail gaps if event hook fails — CourseAttemptHistory write and reset must be atomic	Lost audit data; compliance risk	Wrap snapshot + reset in a database transaction. If the snapshot fails, the reset does not proceed.